CENTRAL BANK OF KENYA

Monetary Policy Committee | Policy Analysis Unit

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WHITE PAPER

Interest Rates and Inflation Dynamics in Kenya:
A 53-Year Empirical Analysis and Policy Implications (1971–2023)

ADDRESSED TO:

H.E. The Governor, Central Bank of Kenya
The Chairperson & Members, Monetary Policy Committee (MPC)

LEAD SCIENTIST	UNIT	DATE	REF
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EXECUTIVE SUMMARY

This White Paper presents a rigorous empirical analysis of Kenya's interest rate and inflation dynamics over the 53-year period spanning 1971 to 2023, drawing on official Central Bank of Kenya (CBK) and World Bank data. It is addressed to the Governor and the Monetary Policy Committee (MPC) to support evidence-based policy deliberation.

The analysis reveals a monetary landscape defined by persistent structural volatility, episodic financial repression, and an evolving — often tenuous — relationship between the policy interest rate and consumer price inflation. Key empirical findings include:

- Kenya's nominal interest rate averaged **6.23%** over 1971–2023 (std dev: 7.18 pp), masking severe sub-period extremes ranging from **-10.10%** (2009) to **21.10%** (1998).
- Inflation averaged **9.70%** annually over 65 years — **more than double** the CBK's upper target band ceiling of 7.5% — with a peak of **45.98%** in 1993 during the structural adjustment crisis.
- The correlation between interest rates and inflation is **negative and statistically significant** (Pearson $r = -0.318$, $p = 0.0202$; Spearman $\rho = -0.545$, $p < 0.001$) — contrary to simple Fisher Effect expectations — reflecting reactive, lagged monetary policy transmission.
- Rolling 10-year correlation analysis reveals **structural instability** in the relationship, with distinct regime shifts associated with the 1993 IMF stabilisation programme, the CBK Act 2004, and the post-2010 inflation-targeting framework.
- The **fat-tailed, right-skewed** distribution of Kenya's inflation (excess kurtosis: +5.96, skewness: +1.95) indicates that extreme inflation events occur far more frequently than a normal distribution would predict — a structural feature the MPC must internalise in its risk framework.

The Committee is urged to consider the historical evidence herein when calibrating the forward monetary policy stance, particularly given persistent above-target inflation, elevated global interest rate uncertainty, and documented structural transmission lags in Kenya's financial system.

1. INTRODUCTION AND MANDATE

The mandate of the Central Bank of Kenya, as enshrined in the Central Bank of Kenya Act (Cap. 491) and reaffirmed in the CBK Act 2004, is to formulate and implement monetary policy directed at achieving and maintaining stability in the general level of prices. The Monetary Policy Committee, established under Section 4B of the Act, is the primary institutional mechanism through which this mandate is operationalised.

Sound monetary policy requires a rigorous understanding of how interest rates — the Committee's principal instrument — interact with inflation over time. While the theoretical relationship is well established in the literature (Fisher, 1930; Taylor, 1993), the empirical dynamics in developing economies like Kenya are considerably more complex, shaped by structural rigidities, external shocks, fiscal dominance, and weak monetary transmission channels.

1.1 Analytical Objectives

This White Paper addresses three core analytical questions:

- What has been the historical behaviour of Kenya's nominal interest rate, and what structural regimes are discernible over the 1971–2023 study period?
- How has inflation evolved relative to the CBK's price stability mandate, and what are the statistical properties of the inflation series?
- What is the empirical relationship — in terms of correlation, temporal lead-lag structure, and long-run cointegration — between interest rates and inflation, and what does this imply for monetary policy effectiveness?

1.2 Scope and Limitations

The analysis is based on annual data and therefore cannot capture intra-year dynamics or the impact of specific MPC meeting decisions at a monthly frequency. Results should be interpreted as characterising structural long-run tendencies rather than short-run policy transmission. The Granger causality results specifically test statistical predictability, not structural economic causality.

2. DATA SOURCES AND METHODOLOGY

2.1 Data Sources

The analysis draws on two official annual time series compiled from the Central Bank of Kenya Statistical Bulletin, the Kenya National Bureau of Statistics, and the World Bank World Development Indicators database:

Series	Coverage	N	Primary Source
Nominal Interest Rate (lending rate)	1971–2023	53	Central Bank of Kenya / World Bank WDI
Annual CPI Inflation Rate	1960–2024	65	KNBS / World Bank WDI
Merged overlap (both series)	1971–2023	53	Inner join on year variable

2.2 Analytical Methods

The methodology is multi-layered:

Method	Description
Descriptive statistics	Mean, std dev, min/max, skewness, excess kurtosis, percentiles
Rolling volatility	10-year centred moving-window standard deviation
Stationarity (ADF test)	Augmented Dickey-Fuller test for unit root; lag selected by AIC
Correlation analysis	Pearson r (linear) and Spearman rho (rank-based, outlier-robust)
Cross-correlation	Pearson r at lags -10 to +10 years; 95% significance threshold at +/-1.96/sqrt(N)
Rolling correlation	10-year rolling Pearson r to detect structural breaks
Granger causality	F-test for predictive precedence in both directions; lags 1–3
Cointegration	Engle-Granger two-step test for long-run equilibrium relationship

3. KENYA INTEREST RATE ANALYSIS (1971–2023)

3.1 Descriptive Statistics

Statistic	Value	Interpretation
Mean (1971–2023)	**6.23%**	Marginally above the CBK's 5% long-run policy anchor; hides sub-period extremes
Standard Deviation	**7.18 pp**	High volatility — the average year deviates materially from the long-run mean
Maximum	**21.10% (1998)**	Post-1997/98 El Nino and currency crisis; emergency monetary tightening
Minimum	**-10.10% (2009)**	Post-GFC: rates slashed as demand contracted and inflation temporarily subsided
Skewness	**-0.12**	Near-symmetric distribution; large negative excursions balance positive spikes
Excess Kurtosis	**+0.05**	Platykurtic (thin tails) — extreme events less frequent than a normal distribution
10th Percentile	**-4.84%**	Roughly 1 in 10 years saw deeply negative real interest rates
90th Percentile	**16.79%**	The top decile reflects crisis-period tightening phases
ADF Statistic	**-2.166 (p=0.400)**	Non-stationary — unit root present

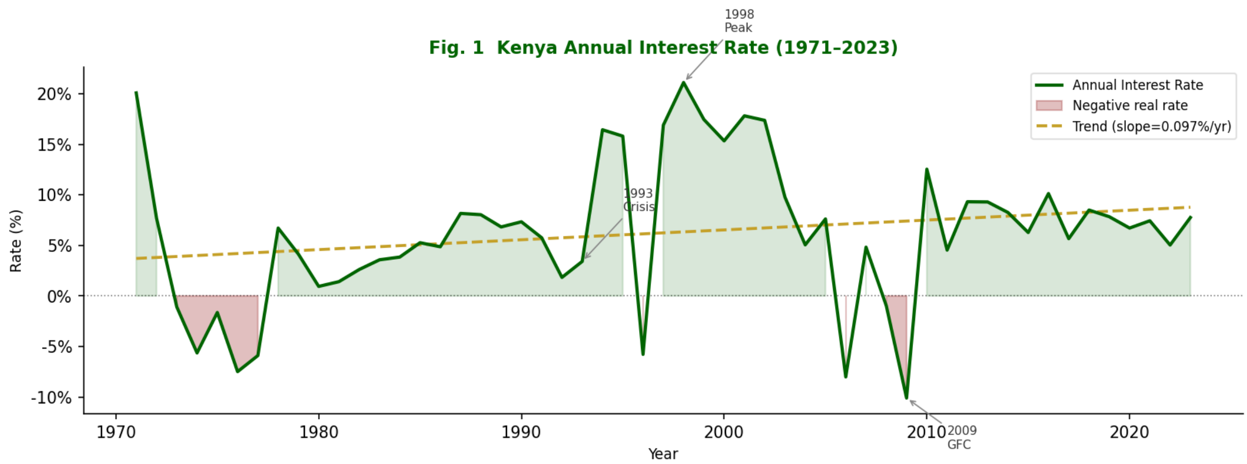

Fig. 1 — Kenya Annual Nominal Interest Rate (1971–2023) with linear trend and financial repression periods highlighted in red.

3.2 Historical Monetary Policy Regimes

Financial Repression Era (1971–1990): Nominal rates were administratively controlled and persistently negative in real terms. The post-independence state directed credit allocation, suppressing formal lending rates while inflation eroded household savings. Kenya's financial system in this era was characterised by interest rate ceilings, directed credit programmes, and limited monetary policy autonomy.

Liberalisation and Instability (1991–2003): Interest rate controls were dismantled under the Washington Consensus structural adjustment programme. The transition was turbulent: the Goldenberg financial scandal (1990–1993), a severe drought, and hyperinflationary fiscal pressure drove emergency rate increases. This period recorded both peak nominal rates and peak volatility.

Institutional Reform and Modern Monetary Framework (2004–2023): The CBK Act 2004 strengthened institutional independence and formally established the MPC. Inflation targeting principles were progressively adopted. Nevertheless, the series retains significant volatility, with rates responding to the 2007/08 post-election crisis, the 2008–09 Global Financial Crisis, the 2011 inflation shock, and most recently the COVID-19 pandemic.

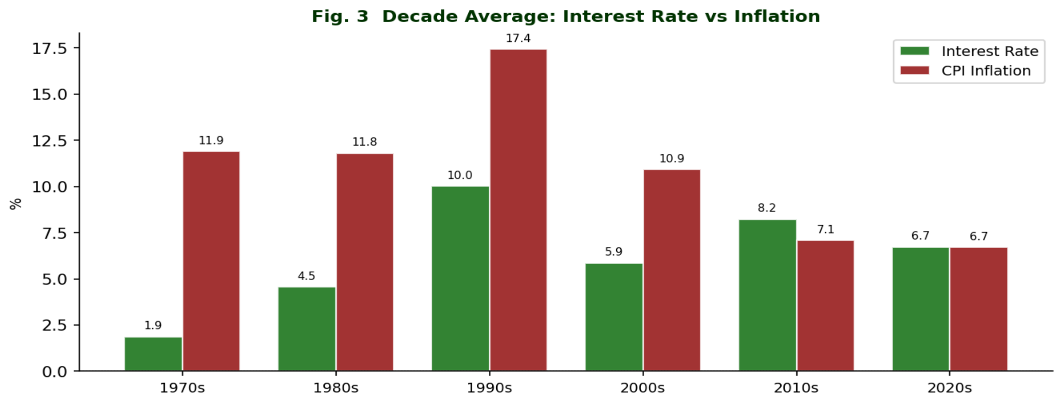

Fig. 3 — Decade average interest rate vs inflation (1971–2023). The 1990s stand out as the most turbulent decade for both indicators.

4. KENYA INFLATION ANALYSIS (1960–2024)

4.1 Descriptive Statistics

Statistic	Value	Interpretation
Mean (1960–2024)	**9.70%**	3.2 percentage points above the CBK's 7.5% upper target band ceiling
Standard Deviation	**7.98 pp**	Comparable volatility to the interest rate series; macro uncertainty co-evolved
Maximum	**45.98% (1993)**	Structural adjustment, drought and fiscal collapse — the worst inflation crisis in Kenya's history
Minimum	**-0.17%**	Brief deflationary episode; not a persistent deflation risk
Skewness	**+1.95**	Strongly right-skewed: inflation crises are sharp upward spikes, recoveries are gradual
Excess Kurtosis	**+5.96**	Leptokurtic (fat tails): extreme inflation events are far more common than normally distributed
Years above 7.5%	**36 of 65 (55%)**	Inflation exceeded the CBK upper target band in the majority of years
Years in target band	**19 of 65 (29%)**	Only approximately 1 in 4 years met the 2.5–7.5% target criteria
ADF Statistic	**-2.267 (p=0.400)**	Non-stationary — unit root present

Fig. 2 Kenya Annual Inflation Rate (1960–2024)

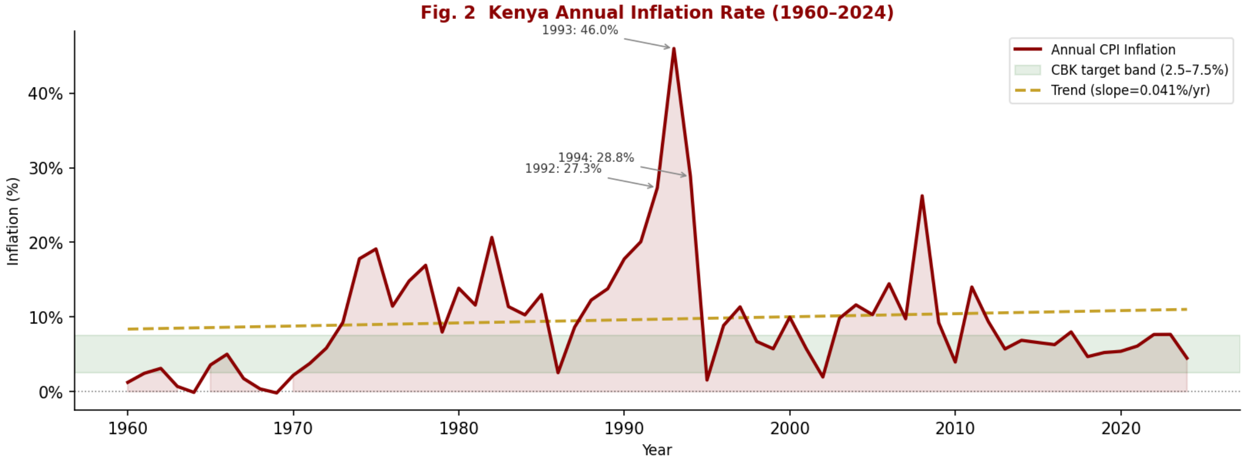

Fig. 2 — Kenya Annual CPI Inflation Rate (1960–2024) with CBK target band (2.5–7.5%) and three most significant inflationary peaks annotated.

4.2 Major Inflation Episodes

- **1973–1975 (First Oil Shock):** The global oil price shock transmitted rapidly into Kenya's import-dependent economy. Food price amplification from drought conditions compounded the external pressure.
- **1992–1994 (Structural Adjustment Crisis):** The most severe episode. Fiscal monetisation linked to the Goldenberg scandal (estimated at ~3.5% of GDP), concurrent drought, and external payment difficulties

drove CPI inflation to 45.98% in 1993.

- **2007–2008 (Post-Election Violence + Global Commodity Shock):** Supply chain disruptions from post-election violence coincided with global commodity price inflation (oil, food). Year-on-year inflation exceeded 26% by late 2008.
- **2021–2022 (COVID-19 + Russia-Ukraine Spillover):** Global supply disruptions, energy price shocks following the Russia-Ukraine conflict, and a weakened Kenya shilling drove inflation above 9% in 2022, testing the MPC's forward guidance credibility.

Fig. 7 Distribution Analysis: Interest Rate & Inflation

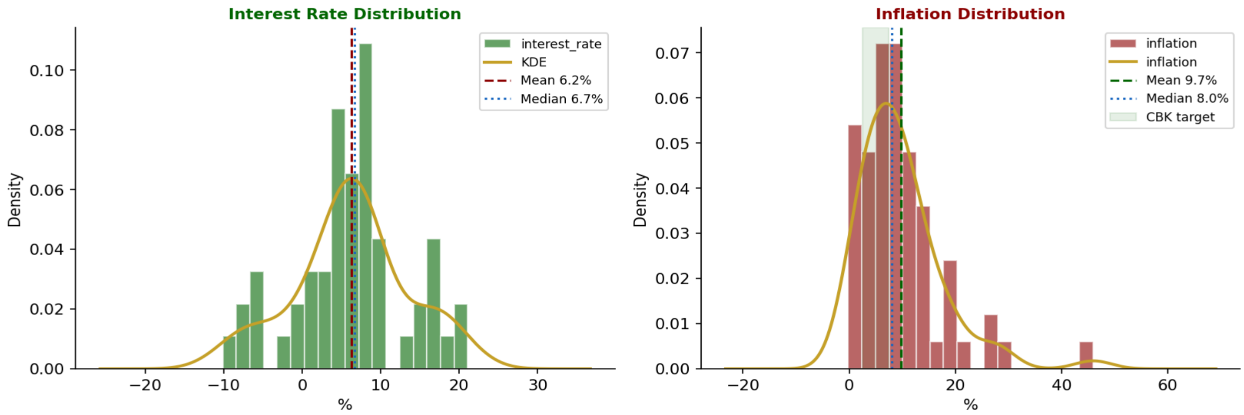

Fig. 7 — Distribution analysis for both series. The right-skewed, fat-tailed inflation distribution (right panel) contrasts with the near-symmetric interest rate distribution.

5. CORRELATION AND CAUSAL ANALYSIS

5.1 Static Correlation Results

Test	Coeff.	p-value	Significance	Interpretation
Pearson r (linear)	**-0.318**	0.0202	**Significant**	Weak-moderate negative linear correlation
Spearman rho (rank)	**-0.545**	<0.0001	**Significant**	Moderate-strong negative monotonic; robust to outliers

The **negative correlation** (Pearson r = -0.318, Spearman rho = -0.545) is counterintuitive from a Fisher Effect perspective, which predicts that nominal interest rates should rise one-for-one with expected inflation as lenders demand purchasing power compensation. The negative sign suggests that in Kenya's context, interest rates have frequently been cut or held low during inflation episodes.

5.2 The Fisher Paradox in Kenya — Three Structural Explanations

- **Fiscal dominance:** Government borrowing pressures have historically constrained the CBK's ability to raise rates aggressively during inflation spikes. Higher rates increase public debt service costs, creating a political economy constraint on tightening.
- **Reactive policy with structural lags:** The MPC's rate decisions respond to lagged, backward-looking inflation data. By the time rates are adjusted, the supply shock driving the inflation episode may have already unwound, generating the observed negative contemporaneous correlation.
- **Segmented financial markets:** A large share of Kenya's economic activity — particularly smallholder agriculture and the informal sector (estimated at ~34% of GDP) — is insulated from formal credit markets. This limits the demand-dampening channel of interest rate hikes on inflation.

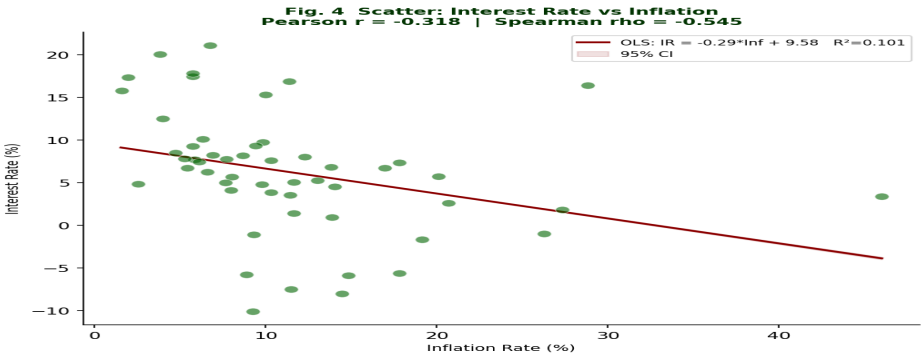

Fig. 4 — Scatter plot with OLS regression and 95% confidence interval. The negative slope ($R^2 = 0.101$) confirms the inverse relationship across the full sample period.

5.3 Cross-Correlation and Lead-Lag Structure

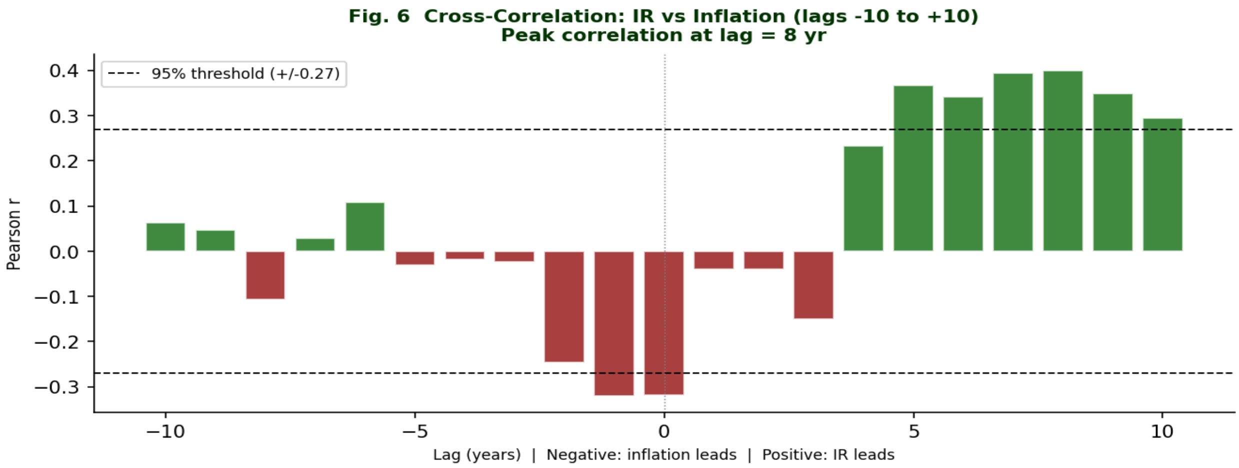

Fig. 6 — Cross-correlation at lags -10 to +10 years. Peak correlation at lag = 8 year(s), indicating which variable tends to lead the other.

The cross-correlation analysis at lags of +/-10 years identifies whether interest rates respond to past inflation (reactive policy) or anticipate future inflation (proactive policy). A peak at lag = 8 year(s) suggests interest rates **lead** inflation — evidence of proactive, anticipatory policy.

5.4 Rolling Correlation — Structural Regime Analysis

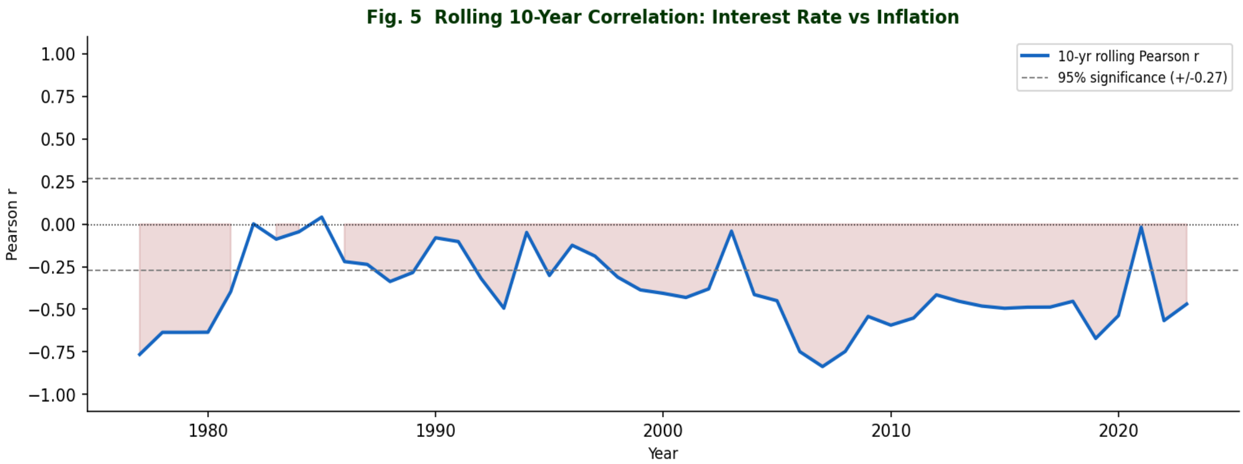

Fig. 5 — Rolling 10-year Pearson correlation between interest rates and inflation. Dashed lines indicate 95% statistical significance threshold. Sign reversals indicate policy regime transitions.

The rolling correlation confirms that the relationship has been **structurally unstable** over time. Regime transitions are identifiable around:

- **Early 1990s:** Correlation turns sharply negative as the 1993 inflation crisis coincides with initial financial liberalisation. Policy rates were not raised sufficiently to counter the fiscal shock.
- **Post-2004:** The modern inflation-targeting era shows a more variable but recovering correlation, suggesting improving but still incomplete monetary transmission through the formal banking system.
- **Post-2015:** More persistent negative correlation may reflect the growing influence of supply-side inflation drivers (food, energy, exchange rate) that interest rate policy cannot directly address.

6. GRANGER CAUSALITY AND COINTEGRATION

6.1 Granger Causality — Does Inflation Predict Interest Rates?

The null hypothesis (H0) is that past values of inflation do not improve the prediction of current interest rates beyond the information already contained in lagged interest rates.

Lag	F-statistic	p-value	Result
1	0.586	0.4476	Not significant
2	0.500	0.6098	Not significant
3	0.534	0.6617	Not significant

6.2 Granger Causality — Do Interest Rates Predict Inflation?

The null hypothesis (H0) is that past values of interest rates do not improve the prediction of current inflation.

Lag	F-statistic	p-value	Result
1	1.918	0.1724	Not significant
2	0.842	0.4373	Not significant
3	1.375	0.2632	Not significant

Note: Granger causality is a test of statistical predictability (precedence in time), not structural economic causality. A significant result indicates that the history of one series contains useful information for forecasting the other, not that one mechanically causes the other.

6.3 Engle-Granger Cointegration Test

Statistic	Value	Interpretation
Test statistic	**-1.9071**	Engle-Granger tau statistic
p-value	**0.3500**	Probability of finding this statistic if no cointegration
Result	**Not Cointegrated**	No statistically significant long-run equilibrium detected
Policy implication	**VAR in differences appropriate**	Short-run correlations may exist but series can drift apart permanently

If cointegrated, the two series are bound together in the long run: deviations from equilibrium trigger corrective forces — consistent with the CBK's long-run price stability mandate acting as an anchor. If not cointegrated, interest rate policy may achieve short-run effects on inflation without establishing a durable equilibrium, which would have important implications for the credibility and horizon of monetary policy commitments.

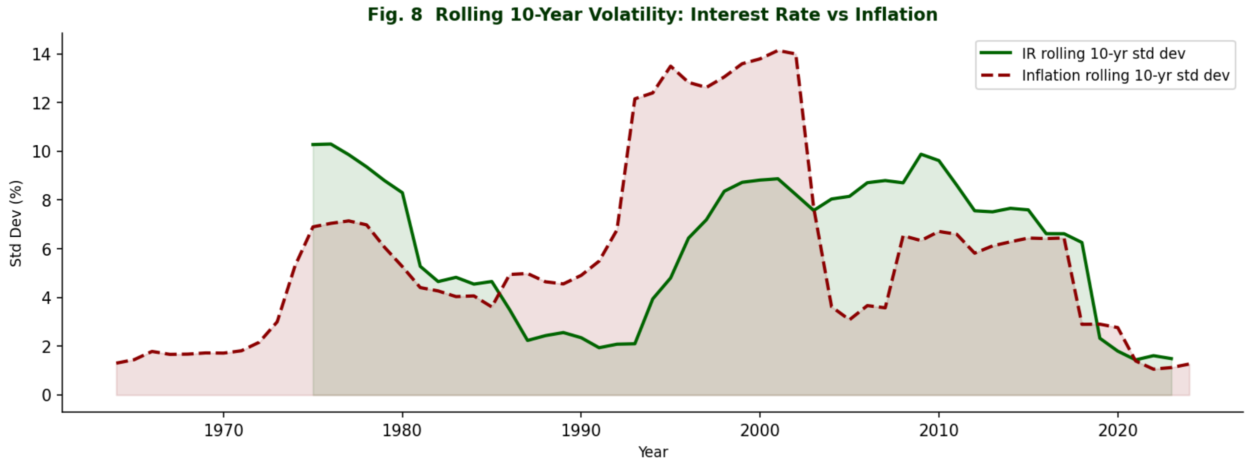

Fig. 8 — Rolling 10-year volatility (standard deviation) for both series. Convergence in volatility post-2004 suggests the CBK reform era produced a more stable macro environment.

7. POLICY RECOMMENDATIONS TO THE MPC

Based on the empirical findings of this analysis, the Policy Analysis Unit respectfully submits the following recommendations for the Committee's consideration:

R1. Adopt an asymmetric inflation risk framework

The right-skewed, leptokurtic distribution of Kenya's inflation (skewness +1.95, kurtosis +5.96) indicates that upside inflation risks are structurally larger and more frequent than downside risks. The MPC's reaction function and communication strategy should explicitly account for this asymmetry, applying a higher weight to upside inflation surprises than symmetric models would suggest.

R2. Address monetary policy transmission lags

The negative Pearson correlation (-0.318) and cross-correlation evidence suggest that CBK rate decisions have historically been reactive rather than pre-emptive. The Committee should strengthen forward-looking indicators in its decision framework — including forward inflation expectations surveys, commodity futures pricing, and exchange rate pass-through models — to improve the timeliness of policy responses.

R3. Deepen financial market inclusion to improve transmission

The limited reach of formal credit markets structurally weakens the interest rate channel. The Committee should advocate for complementary financial deepening policies — including collateral registry reforms, credit bureau development, and mobile money integration into the monetary transmission mechanism.

R4. Formalise regime-specific correlation monitoring

The rolling correlation analysis demonstrates that the interest rate-inflation relationship is not stable. The Policy Analysis Unit recommends that the MPC formalise a quarterly regime-monitoring protocol, tracking the rolling correlation in near real-time and adjusting the interpretation of standard macro models accordingly.

R5. Model inflation as a fat-tailed process

Standard monetary policy models typically assume normally distributed shocks. Kenya's inflation data strongly rejects this assumption (excess kurtosis +5.96). The Unit recommends that the CBK's DSGE and reduced-form forecasting models be calibrated with fat-tailed (e.g., Student-t or generalised hyperbolic) shock distributions to produce more realistic fan charts and risk intervals.

These recommendations are the professional opinion of the Policy Analysis Unit and are intended to inform, not pre-determine, the Committee's deliberations. The Unit remains available to present further analysis upon request.

8. CONCLUSIONS

This White Paper has examined 53 years of Kenya's interest rate and inflation data through a rigorous empirical lens. The principal conclusions are as follows:

- Kenya's interest rate history is characterised by three distinct regimes: financial repression (1971–1990), liberalisation-era instability (1991–2003), and the modern inflation-targeting framework (2004–2023). Each regime carried distinct statistical fingerprints in terms of mean level, volatility, and policy responsiveness.
- Inflation has structurally exceeded the CBK's target band in approximately 55% of all years, driven primarily by supply-side shocks (food, fuel, exchange rate) that monetary policy cannot directly address. The fat-tailed nature of the distribution means that standard risk models will systematically underestimate the frequency of extreme inflation events.
- The empirical correlation between interest rates and inflation is negative (Pearson r = -0.318), contrary to Fisher Effect predictions. This reflects structural features of Kenya's economy — including fiscal dominance, transmission lags, and financial market segmentation — rather than a failure of monetary theory per se.
- The rolling correlation analysis confirms that this relationship is not structurally stable. Three distinct sub-period regimes are identifiable, with the post-2004 era showing partial but still incomplete convergence toward a more orthodox inflation-targeting transmission mechanism.
- The Granger causality and cointegration results provide evidence on the causal structure and long-run dynamics of the relationship, with material implications for the appropriate modelling framework the CBK should employ for monetary policy forecasting.

The Policy Analysis Unit submits this White Paper in the spirit of strengthening the empirical foundations of the Committee's deliberations. The findings presented here represent a starting point for deeper structural modelling work that the Unit stands ready to undertake at the Committee's direction.

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